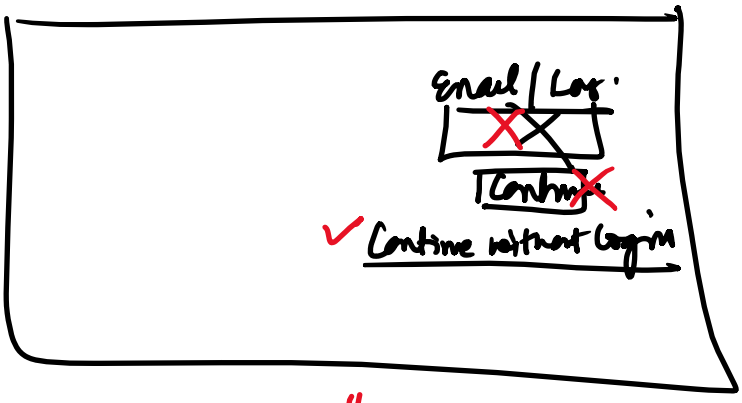


"Generative AI" → Create a new content – Text, code, image, video, audio.

↳ Powered By "FM / LLM"

FM → Pretrained Model which is trained on Internet scale of Data (Raw Data)

↓

Subset of FM ⇒ "LLM" (Natural language processing)

↓

"Text"

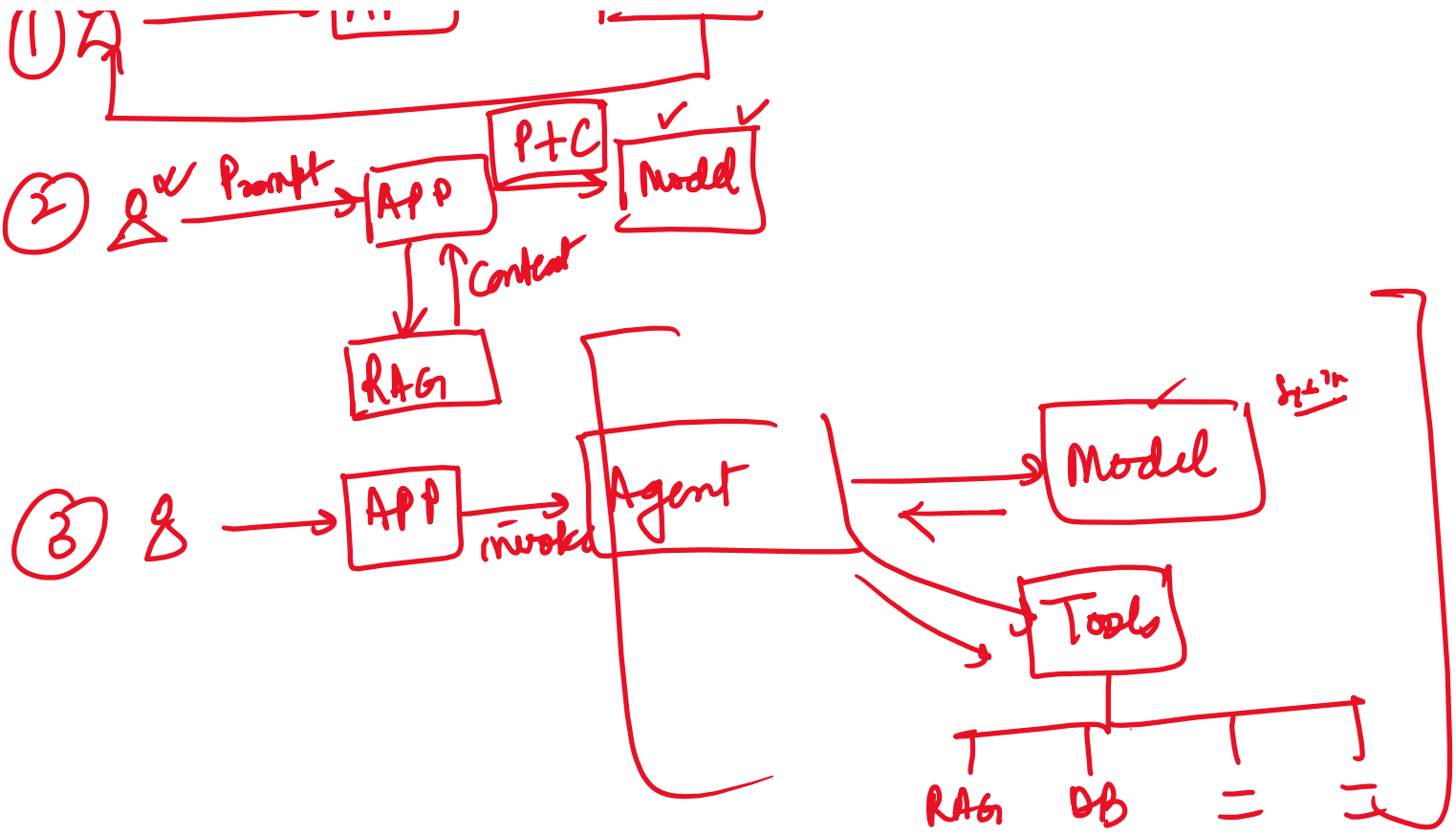
FM / LLM → "Context window"

↑

App logic / open source framework (Langchain, Strands, CrewAI)

Agentic AI System





AgentCore Components

1. Runtime (Env + Framework)
2. Memory (Short-term + Long-term)
3. Gateway (Tools)
4. Identity (who? what?)
5. Observability (Monitoring)

Development

- LLM = support/provide Reasoning capabilities
- Framework → orchestrates the agent's workflow (open-source)

AgentCore → managed Infrastructure & supporting services for production operations.

AgentCore → ... for Production operations.

[Agentic AI App = Strands framework + Claude Opus.

↳ Deploy → [AgentCore Runtime + connect enterprise API's (AgentCore Gateway) + enable sessions + preserve the details (Memory) monitoring → (AgentCore observability)

To secure (AgentCore Identity) → who? what?
Inbound Bind Outbound

1. Runtime ✓ → [Compute, env] + Framework
2. Identity ✓ → who can get accss to agent? → Inbound
   what an agent can get accss to? → outbound
3. [Gateway ✓ → [Tools | Resources]
   [Agent (MCP client)] → Gateway (MCP serv) — API / Lambda / =
4. [Memory
   - Short-term = sessions [current conversation
   - Long-term = Preserve the info → Built-in namespaces → [summary, preference, semantic
5. Observability ⇒ Monitoring
   [what agent(s) doing?
6. Evaluations ⇒ "Agent" behavior?

(6) Evaluations ⇒ Agent behavior

↳ Built-in → metrics | evaluation criteria.

Open-source framework

↳ [Define HOW the agent behaves.]

↘ Code → Define Agent

Amazon Bedrock

↑

Managed Service APIs

Runtime → Provide the env → agent execute.

Demo

Secure AI Agent Access with Amazon Bedrock Identity

(1) Build & Test the Agent → Artifact.

(2) Deploy the Agent to AgentCore Runtime.

+

⇒ [AgentCore Identity – Inbound] + [Cognito user]

(3) Invoke the Deployed Agent

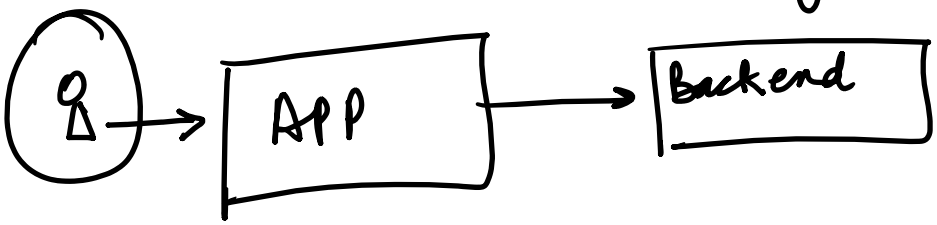
Demo → Amazon Gateway (Tools)

↳ MCP

Lg MCP

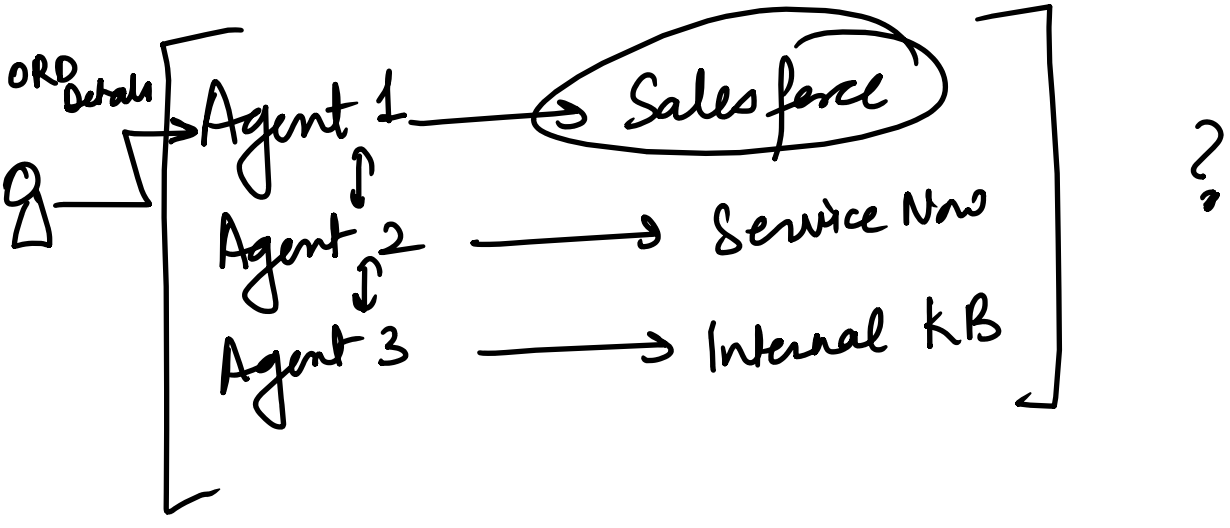
Lg A2A

Amazon Memory — Short-term, Long-term.. — 30 mins



Cust
emp
Apps
Batch jobs
multiple Agents

Tools
APIs
External servs



?

CHALLENGES

(1) Accesscontrol & delegation.
(2) Identity Lifecycle Management
(3) 3rd party = Salesforce, ServiceNow.

③ 3rd party = Salesforce, ServiceNow

④ Governance & Compliance

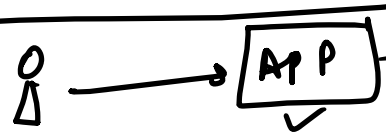
* Inbound Auth → Who is allowed to access / invoke an agent?

Outbound Auth → How does the agent authenticate when accessing the system?

A Inbound Auth : THREE (Common)

"Serverless"

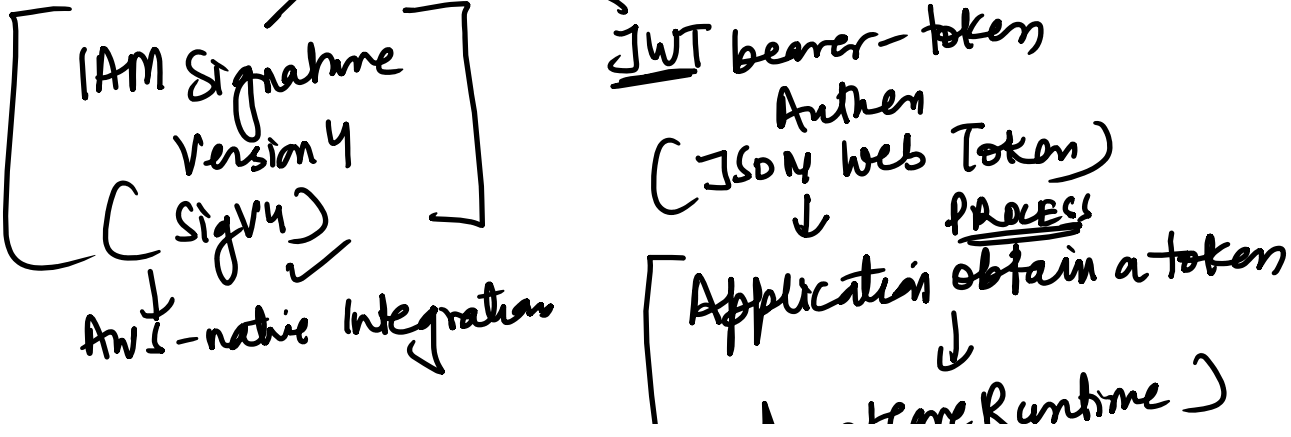
① User authentication



② API key Access : Sys-to-Sys Integration
↳ IAM SigV4 / JWT token

③ Federated Auth

Amazon Bedrock AgentCore Runtime (Inbound Auth)



AWS-native "..."

↓
AgentCore Runtime)
↓
Runtime validates the token

Outbound Auth (Agent → Tools / systems)

① AWS IAM Roles

Agent → Runtime → execution roles → S3 Bucket

② Machine-to-Machine (OAuth 2.0 client credentials)

③ user-delegated acces (OAuth 2.0 Authorization code flow

Demo: AgentCore Runtime & Identity

[AI Helpdesk Agent → Employees]

↳ security gap in this Prototype

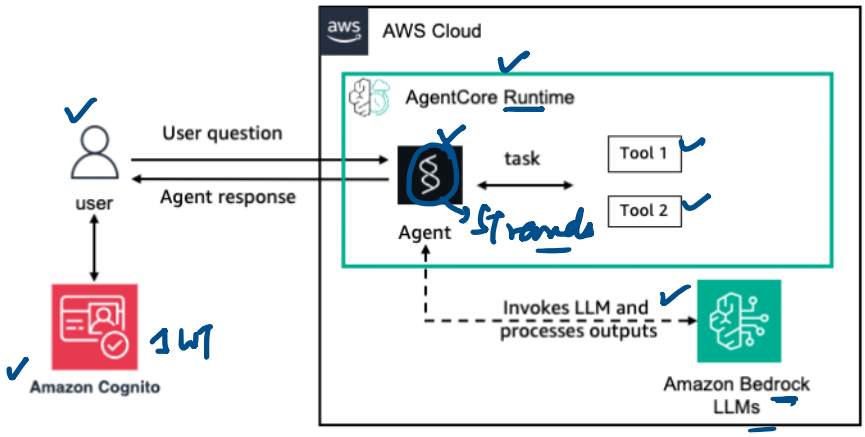
↳ It doesn't verify the identity of Employee behind each Request.

① Deploy an agent to Amazon Bedrock AgentCore Runtime with AgentCore Identity - Inbound Auth

② Sign in to Cognito user pool, get JWT, decode it

③ Rejects callers without valid JWT.

Runtime
Identity
Cognito
Bedrock (LLM)



Memory Types : SHORT-term & Long-term

↓ Raw Interaction (Conversation / current session, agent state checkpoint)

↓ Extract Meaningful Info

~~Long term~~

(1) ~~user preferences~~

Current challenges : [Agentic Memory.]

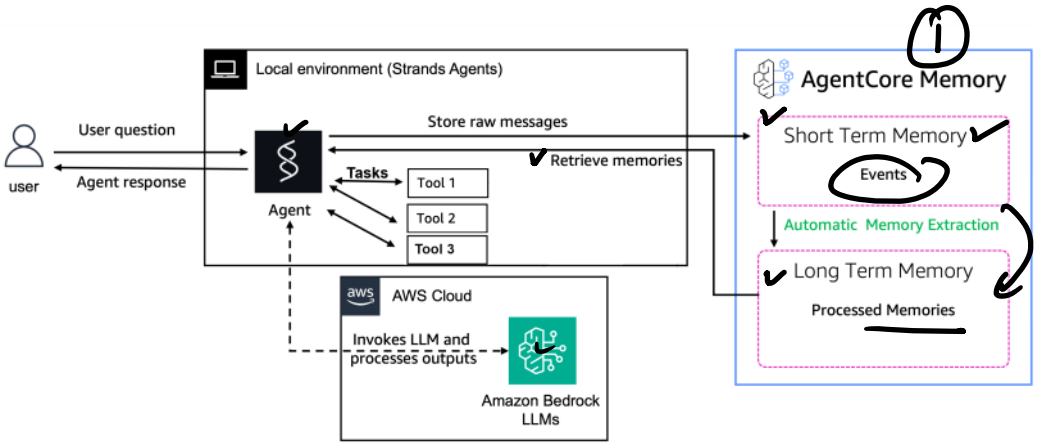
(1) Storage scalability
(2) Memory Refresh (90 days)
(3) Retrieved accuracy
(3) Integration complexity.
(4) Privacy & security.

Demo: Build a Context-Aware Agents — AgentCore Memory

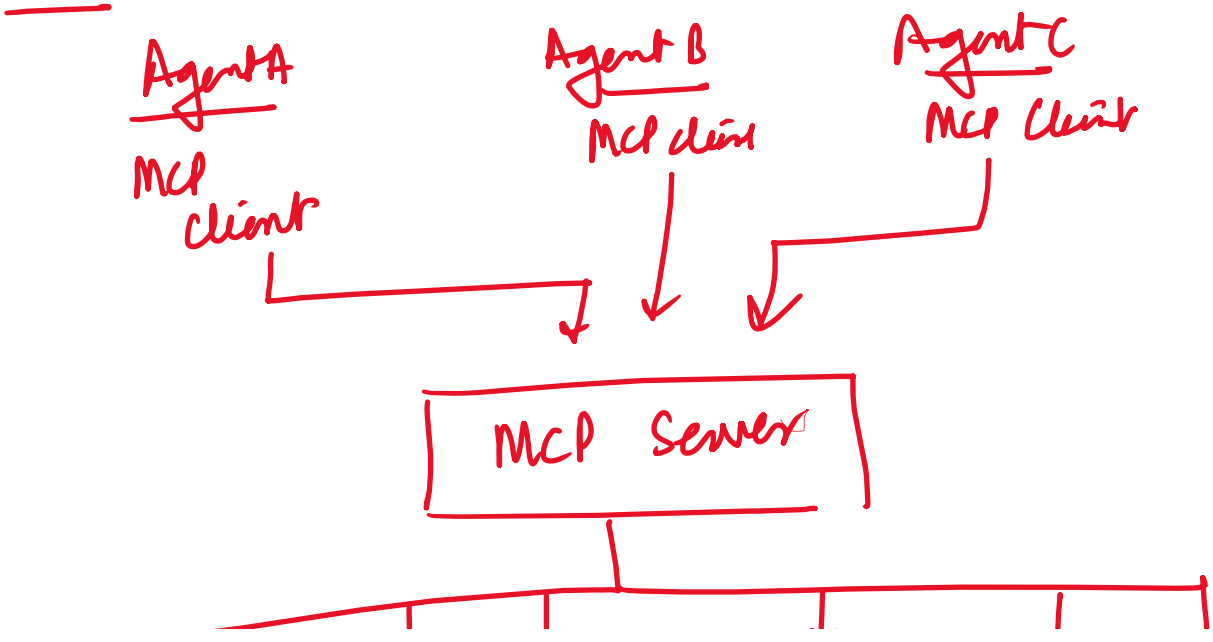
(1) Create AgentCore Memory → Long-term
↳ Semantic, user P, Summary

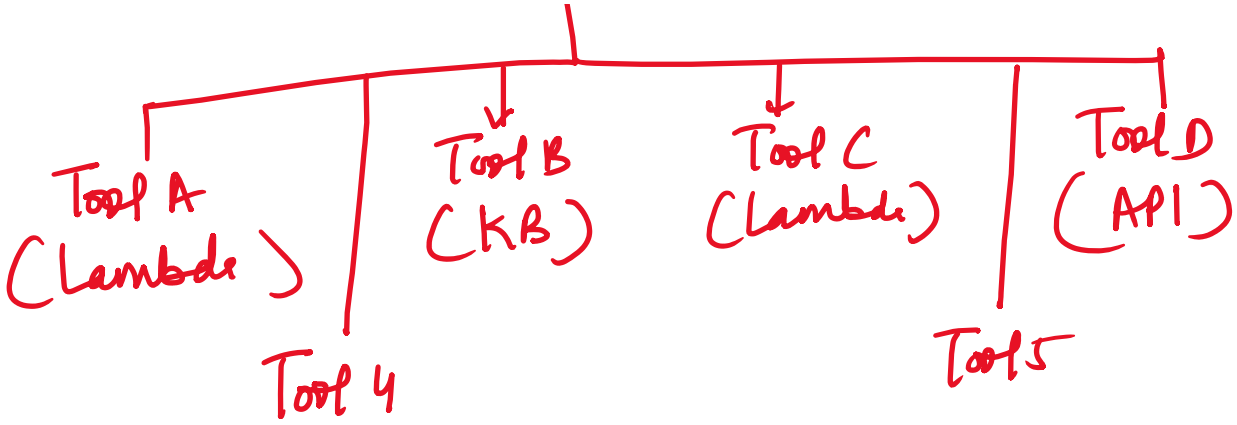
(2) Integrate persistent memory into Strands agent & Configure Long-term retrieval

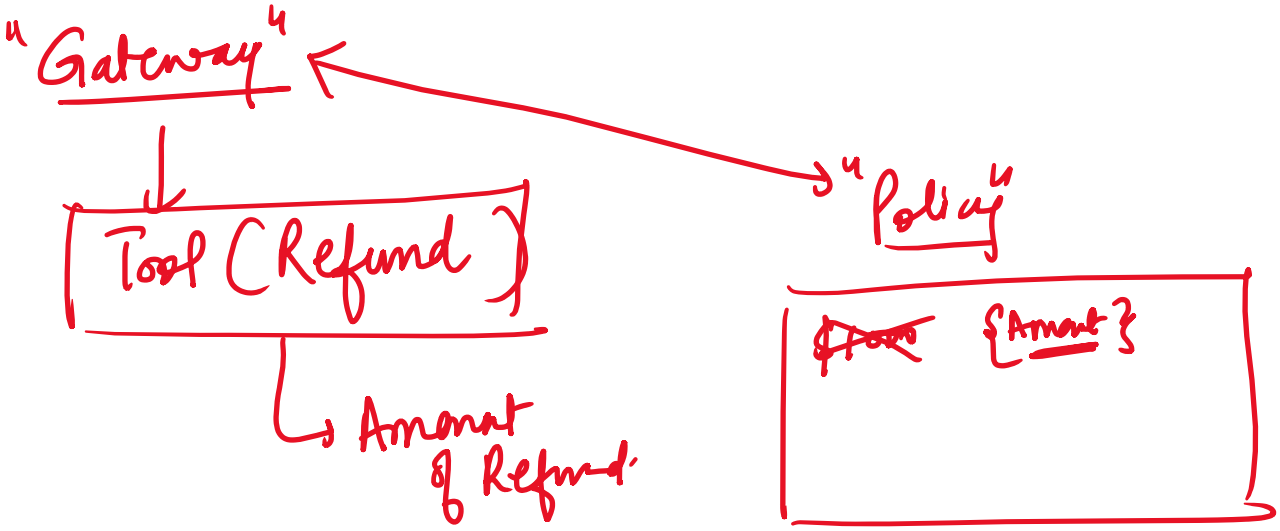
(3) Inspect stored memories → AgentCore Memory



MCP




Tool A
(Lambda)
Tool 4
Tool B
(KB)
Tool C
(Lambda)
Tool 5
Tool D
(API)


"Gateway"
Tool (Refund)
Amount of Refund
"Policy"
{Amount}